

- ① A Security Researcher publicly releases working exploit for a vulnerability before vendor has issued a patch, arguing that public pressure is only way to force a fix. Discuss whether this qualifies as ethical hacking & whether this person falls on the hacker spectrum or not?

Example Scenario:-

Suppose a Security researcher discovers a vulnerability in an online banking application that allows attackers to gain unauthorised access to customer accounts. The researcher reports the issue to software company, but the company does not release a patch for several months. To pressure the company into taking action, the researcher publicly releases the vulnerability details along with a working exploit.

Is this ethical hacking:-

There are different opinions on whether this action is considered ethical hacking.

Arguments: (in Favour)

- The researcher's intention is to improve security rather than causing harm.

- public disclosure increases awareness of vulnerability.
- It encourages software vendors to develop & release a patch more quickly.
- organisations can implement temporary security measures while waiting for official fix.

Arguments: (Against)

- Releasing a working exploit before a patch is available puts users at a risk.
- Cybercriminals can use exploit to attack vulnerable systems.
- Ethical hackers are generally expected to follow responsible disclosure by giving vendor sufficient time to fix the vulnerability before making technical details public.

Hacker Spectrum Classification:

Based on the Scenario, the Researcher is best classified as a "Grey Hat Hacker"

White Hat Hacker:

A Hacker who works with permission, follows ethical guidelines & responsibly reports vulnerability.

ities without exposing users to unnecessary risks.

Grey Hat Hacker:

Who has good intentions but may ignore accepted ethical practices, such as releasing vulnerability details before a patch is available. In this scenario, the researcher is trying to improve security but increases the risk to users by publishing a working exploit.

Black Hat Hacker:

Intentionally exploits vulnerabilities for personal gain, financial profit or to cause damage. This does not match researcher's intentions.

Conclusion:

The researcher aims to improve cybersecurity by forcing the vendor to fix the vulnerability, which reflects a positive intention. However, releasing a working exploit before a security patch is available increases the risk of cyber attacks & may expose many users to harm.

Therefore, The researcher is generally considered as Grey Hat Hacker rather than a White Hat Hacker.